Mass–Energy at Work: Collisions, Decays, and Thresholds

PHYS 2203 — Introductory Physics III

October 12, 2026

Overview

Lectures 17 and 18 developed relativistic momentum and energy, the relation $E^2 = p^2c^2 + m^2c^4$ that connects them. This lecture uses those results in representative calculations. We begin with the invariant mass of a system and use it in two problems: an inelastic collision, in which kinetic energy becomes mass, and the minimum energy needed to create an antiproton in a collision. We then consider particles with no mass, for which $E = pc$, and the decay of a pion at rest into a muon and a massless neutrino, in which mass becomes kinetic energy.

Recall from Lecture 18. For a particle of mass m and speed u, with $\gamma = 1/\sqrt{1 - u^2/c^2}$,

$$\mathbf{p} = \gamma\, m\mathbf{u}, \qquad E = \gamma\, mc^2 = K + mc^2, \qquad K = (\gamma - 1)\, mc^2, \qquad E^2 = p^2c^2 + m^2c^4, \tag{1}$$

and for an isolated system both $\sum_i \mathbf{p}_i$ and $\sum_i E_i$ are conserved, but $\sum_i m_i$ is not.

Units. In particle physics we measure energy in MeV, momentum in MeV/c, and mass in MeV/c^2. Then $E^2 = (pc)^2 + (mc^2)^2$ is just arithmetic in MeV, with no SI conversions.

1 The Invariant Mass of a System

For a single particle, Lecture 18 showed that $E^2 - p^2c^2 = m^2c^4$ is the same in every frame. The same construction works for a collection of particles.

Invariant mass of a system. For a system of particles with total energy $\sum_i E_i$ and total momentum $\sum_i \mathbf{p}_i$,

$$M_{\text{sys}}^2 c^4 \;\equiv\; \left(\sum_i E_i\right)^2 - \left(\sum_i \mathbf{p}_i\right)^2 c^2.$$

It has the same value in every inertial frame, and it is *conserved*, because $\sum_i E_i$ and $\sum_i \mathbf{p}_i$ separately are. In general it is *not* the sum of the masses of the particles.

2 Kinetic Energy Becomes Mass: An Inelastic Collision

Two identical particles of mass m approach each other with equal and opposite speeds u, collide, and stick together to form a single object of mass M. What is M?

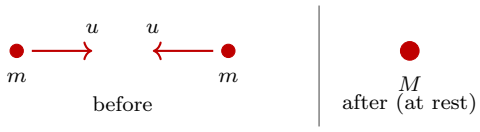


Figure 1: Two masses m collide head-on with equal speeds u and merge into a single mass M.

Momentum. By symmetry the total momentum is zero before, $\gamma mu - \gamma mu = 0$, so it is zero after: the merged object is at rest.

Energy. Before, each particle has $E = \gamma mc^2$. After, the merged object is at rest with $E = Mc^2$. Conserving total energy,

$$2\gamma\, mc^2 = Mc^2 \qquad \Longrightarrow \qquad \boxed{M = 2\gamma\, m > 2m.} \tag{2}$$

The merged object is heavier than its parts. The mass gained in the collision is

$$\Delta M = M - 2m = 2\gamma m - 2m = 2m(\gamma - 1). \tag{3}$$

Now compare with the kinetic energy. Before the collision, each particle has kinetic energy $K = (\gamma - 1)mc^2$, so the two together have $2(\gamma - 1)mc^2$. After the collision the merged object is at rest, so its kinetic energy is zero. The change in kinetic energy is therefore

$$\Delta K = K_{\text{after}} - K_{\text{before}} = 0 - 2(\gamma - 1)mc^2 = -2mc^2(\gamma - 1). \tag{4}$$

It is negative because kinetic energy is lost. Comparing the two results,

$$\boxed{\Delta M = -\frac{\Delta K}{c^2}.} \tag{5}$$

The kinetic energy has not been lost. It is now rest energy of the merged object, stored as its internal (thermal) energy. For $u = 0.8c$, $\gamma = 5/3$ and $M = 3.33\, m$: two-thirds of the product's mass is former kinetic energy.

Does every observer agree? The masses m and M are frame-independent, so ΔM is the same in every frame. Because total energy is conserved, $\Delta K = -\Delta M\, c^2$ is the same in every frame too, even though each observer measures different kinetic energies before and after. Checking this directly in a frame moving with one of the particles takes velocity addition and two different γ factors. The invariant mass avoids that work. Before the collision $M_{\text{sys}}c^2 = 2\gamma mc^2$, and after it the system is a single object, so $M_{\text{sys}} = M$. Since M_{sys} is conserved and the same in every frame, $M = 2\gamma m$ for every observer.

3 Threshold Energy: Making Antimatter

To find the minimum energy needed to create new particles, we use the invariant mass from Section 1. In the frame where the total momentum is zero, called the **centre-of-momentum (CM) frame**, its definition reduces to

$$M_{\text{sys}}c^2 = \Big(\sum_i E_i\Big)_{\text{CM}}. \tag{6}$$

This is the energy available to create new particles, the energy not tied up in the motion of the system as a whole.

The CM frame is the frame of an observer moving along with the system as a whole. To find how fast it moves, we need a relation between a particle's speed and its energy and momentum. Divide the momentum $p = \gamma m u$ by the energy $E = \gamma m c^2$. The factors of γ and m cancel:

$$\frac{p}{E} = \frac{\gamma\, m u}{\gamma\, m c^2} = \frac{u}{c^2}. \tag{7}$$

Multiplying both sides by c,

$$\boxed{\frac{u}{c} = \frac{pc}{E}.} \tag{8}$$

Once a particle's energy and momentum are known, this gives its speed directly, without finding γ first. Now treat the whole system as a single object with total energy $\sum_i E_i$ and total momentum $\sum_i \mathbf{p}_i$. Applying Eq. (8) to this single object gives the velocity of the CM frame relative to the lab:

$$\frac{V_{\text{CM}}}{c} = \frac{\left(\sum_i p_i\right) c}{\sum_i E_i}. \tag{9}$$

The inelastic collision of Section 2 was already worked in its CM frame, since the two particles had equal and opposite momenta. In most problems, though, the lab is not the CM frame. We never need to transform quantities into the CM frame. Because M_{sys} has the same value in every frame, we can compute it in the lab frame and interpret it in the CM frame.

Example: the minimum energy to make an antiproton

The question. A beam of protons strikes protons at rest in a target. We wish to create an antiproton, which is a negatively charged partner of a proton. Conservation of charge requires that an extra proton be created along with it:

$$p + p \;\longrightarrow\; p + p + p + \bar{p}. \tag{10}$$

The antiproton has the same mass as the proton, $mc^2 = 938$ MeV. What is the minimum beam energy?

The idea. At *threshold* the reaction is only just possible, so no energy goes into relative motion of the products: in the CM frame all four are produced *at rest*. Evaluate M_{sys} before the collision in the lab frame and after it in the CM frame, and set the two equal.

After, in the CM frame. Four particles at rest, each of mass m:

$$M_{\text{sys}} c^2 = 4mc^2 \qquad \Longrightarrow \qquad M_{\text{sys}}^2 c^4 = 16\, m^2 c^4. \tag{11}$$

Before, in the lab frame. The projectile has total energy E and momentum p, and the target is at rest with energy mc^2 and no momentum:

$$\begin{aligned} M_{\text{sys}}^2 c^4 &= \left(E + mc^2\right)^2 - p^2 c^2 \\ &= E^2 + 2Emc^2 + m^2 c^4 - \left(E^2 - m^2 c^4\right) \\ &= 2Emc^2 + 2m^2 c^4, \end{aligned} \tag{12}$$

where we used $p^2 c^2 = E^2 - m^2 c^4$ for the projectile.

Equate and solve.

$$2Emc^2 + 2m^2c^4 = 16\,m^2c^4 \qquad \Longrightarrow \qquad E = 7\,mc^2. \tag{13}$$

The beam's *kinetic* energy at threshold is

$$\boxed{K = E - mc^2 = 6\,mc^2 = 6\,(938\text{ MeV}) = 5.6\text{ GeV}.} \tag{14}$$

We can check this by finding how the products move in the lab. At threshold the beam proton has $pc = \sqrt{E^2 - m^2c^4} = \sqrt{48}\,mc^2 = 6.93\,mc^2$, and the total energy in the lab is $E + mc^2 = 8\,mc^2$. The CM frame therefore moves relative to the lab at

$$\frac{V_{\text{CM}}}{c} = \frac{6.93}{8} = 0.866. \tag{15}$$

The four products are at rest in the CM frame, so in the lab they all move together at $0.866c$, with $\gamma = 2$. Each product has total energy $\gamma mc^2 = 2mc^2$, so together they have $4(2mc^2) = 8\,mc^2$, matching the total before the collision. Of this, $4mc^2$ is the rest energy of the four particles and $4mc^2$ is their kinetic energy. So of the beam's $6mc^2$ of kinetic energy, only $2mc^2$ becomes the rest energy of the new proton and antiproton. The other $4mc^2$ stays as kinetic energy, because the products must keep moving to carry the beam's momentum.

The two new particles have rest energy $2mc^2 = 1.9$ GeV, yet the beam must supply 5.6 GeV, three times as much. Momentum conservation forces the products to keep moving in the lab, and that motion carries kinetic energy that cannot become mass. Only the CM energy is available.

Colliders and fixed targets. For a fixed target, $M_{\text{sys}}c^2 = \sqrt{2Emc^2 + 2m^2c^4} \approx \sqrt{2Emc^2}$ grows only as $\sqrt{E}$. In a collider, two beams of energy E meet head-on with $\sum_i \mathbf{p}_i = 0$, so the lab *is* the CM frame and $M_{\text{sys}}c^2 = 2E$. At 1 TeV per beam, a collider reaches 2 TeV, while a fixed target reaches only 43 GeV.

The Bevatron at Berkeley was built to reach 6.2 GeV, just above this threshold. Chamberlain and Segrè discovered the antiproton with it in 1955 and shared the 1959 Nobel Prize.

4 Massless Particles

Every formula in Lecture 18 was derived for a particle with mass. What if $m = 0$?

The usual formulas fail. Setting $m = 0$ in $\mathbf{p} = \gamma m\mathbf{u}$ and $E = \gamma mc^2$ gives 0 — unless the particle moves at exactly $u = c$, in which case $\gamma = \infty$ and both become the indeterminate form $0 \times \infty$. They tell us nothing.

But the energy–momentum relation survives. Nothing in

$$E^2 = p^2c^2 + m^2c^4 \tag{16}$$

requires $m \neq 0$. Setting $m = 0$ gives

$$\boxed{E = pc \qquad \text{(massless particle).}} \tag{17}$$

Eq. (8) confirms that such a particle moves at the speed of light: with $E = pc$, it gives $u/c = pc/E = 1$.

Massless particles. A particle with $m = 0$ has $E = pc$ and travels at exactly $u = c$ in every inertial frame. It has no rest frame — consistent with Lecture 14, where we saw that a lightlike interval admits no frame in which the two events occur at the same place, so no clock can ride along. Its energy and momentum are *not* determined by its speed, which is always c. They are independent properties.

A system of massless particles can still have mass. Two photons flying in opposite directions with energy E each have total momentum zero and total energy $2E$, so their invariant mass is $M_{\text{sys}}c^2 = 2E$, even though each photon is massless.

5 A Decay at Rest: The Pion

A positively charged pion ($m_\pi c^2 = 139.6$ MeV) is an unstable particle. At rest, it decays into a muon ($m_\mu c^2 = 105.7$ MeV) and a neutrino, which we treat as massless:

$$\pi^+ \longrightarrow \mu^+ + \nu. \tag{18}$$

Find the energy of the neutrino, and the kinetic energy and speed of the muon.

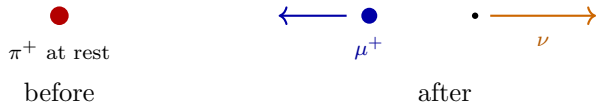


Figure 2: A pion at rest decays into a muon and a neutrino moving in opposite directions.

Momentum. The pion is at rest, so the total momentum is zero. The muon and neutrino must move in opposite directions with momenta of equal magnitude. Call that magnitude p.

Energy. Before the decay the total energy is the pion's rest energy. After, it is shared between the muon and the neutrino, and the neutrino is massless, so $E_\nu = pc$:

$$m_\pi c^2 = E_\mu + pc. \tag{19}$$

Solve. For the muon, $E_\mu^2 = p^2c^2 + m_\mu^2 c^4$. Substitute $pc = m_\pi c^2 - E_\mu$ from the energy equation:

$$E_\mu^2 = \left(m_\pi c^2 - E_\mu\right)^2 + m_\mu^2 c^4 = m_\pi^2 c^4 - 2m_\pi c^2 E_\mu + E_\mu^2 + m_\mu^2 c^4. \tag{20}$$

The E_μ^2 terms cancel, leaving

$$E_\mu = \frac{m_\pi^2 c^4 + m_\mu^2 c^4}{2m_\pi c^2} = \frac{(139.6)^2 + (105.7)^2}{2(139.6)}\ \text{MeV} = 109.8\ \text{MeV}. \tag{21}$$

Then

$$E_\nu = pc = m_\pi c^2 - E_\mu = 139.6 - 109.8 = 29.8\ \text{MeV}, \qquad K_\mu = E_\mu - m_\mu c^2 = 109.8 - 105.7 = 4.1\ \text{MeV}, \tag{22}$$

and the muon's speed is

$$\frac{u}{c} = \frac{pc}{E_\mu} = \frac{29.8}{109.8} = 0.27, \tag{23}$$

from Eq. (8).

The rest energy of the pion is larger than the rest energy of the products by $139.6 - 105.7 = 33.9$ MeV, and this difference appears as kinetic energy: 29.8 MeV for the neutrino and 4.1 MeV for the muon. This is the inelastic collision run in reverse, with mass turning into kinetic energy. Notice that the two particles have the same momentum, but the massless neutrino carries most of the kinetic energy. When a light particle and a heavy particle share momentum equally, the light particle gets most of the kinetic energy.